

Program : Diploma in Automation and Robotics	
Course Code: 6332C	Course Title: Radar and Navigation
Semester: 6	Credits: 4
Course Category : Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the concept of Radar , its applications and different Radar performance factors
- To understand the operation of FM - CW Radar, MTI & Pulse Doppler Radar
- To understand the concept of Navigation and types of radio navigation.
- To understand the two types of landing systems
- To understand satellite navigation systems

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics knowledge of electrical electronic devices		Fundamentals of Electrical and Electronics Engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concept of Radar , its applications and different Radarperformance factors	13	Understanding
CO2	Understand the operation of FM - CW Radar, MTI & Pulse Doppler Radar	15	Understanding
CO3	Understand the concept of Navigation and types of radio navigation	16	Understanding
CO4	Understand the two types of landing systems and satellite navigation systems	14	Understanding
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	2					
CO3			3				
CO4	3						

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concept of Radar , its applications and different Radar performance factors		
M1.01	Describe the block diagram of Radar	2	Understanding
M1.02	State the Radar frequency ranges and derive the radar range equations and block diagram of FM - CW super heterodyne receiver	4	Understanding
M1.03	Explain the operation of MTI Radar and the operation and block diagram of MTI Radar employing power amplifier	4	Understanding
M1.04	Explain the Radar performance factors	3	Understanding
Contents: Radar – concept - block diagram – operation – frequency ranges – applications – range equation – derivation (no problems) - Radar performance factors – minimum detectable signal -receiver noise – Signal to Noise ratio -Radar cross section of targets – transmitter power – pulse repetition frequency – range ambiguities			
CO2	Understand the operation of FM - CW Radar, MTI & Pulse Doppler Radar		
M2.01	Understand Doppler effect with equation (no derivation) and to explain the block diagram of FM - CW Radar	3	Understanding
M2.02	Explain the operation of MTI Radar and the operation and block diagram of MTI Radar employing power amplifier	3	Understanding
M2.03	Explain the operation of a delay line transfer and the block diagram of MTI Signal Processor	2	Understanding

M2.04	Describe the operation of Pulse Doppler Radar	2	Understanding
M2.05	Explain Tracking Radar and list out its types	2	Understanding
M2.06	Explain the types of Radar displays	3	Understanding
	Series Test – I	1	
Contents: Doppler effect - FM - CW Radar - FM - CW Radar with super heterodyne receiver – block diagram - MTI Radar - MTI Radar with power amplifier transmitter -delay line cancelers - MTI Signal Processor - Pulse Doppler Radar - Tracking Radar – basic block diagram – types - Radar displays – A Scope – PPI Scope			
CO3	Understand the concept of Navigation and types of radio navigation		
M3.01	To define the concept of navigation and four methods of navigation	2	Understanding
M3.02	To explain the working of loop antenna and Goniometer	4	Understanding
M3.03	To explain the working principle of ADF / NDB – Radio compass ADF, VHF phase comparison ADF	3	Understanding
M3.04	To explain the working principle of hyperbolic navigation systems – LORAN, Omega, DECCA	4	Understanding
M3.05	To explain the principle of different types of radio ranges - VOR, Doppler VOR, DME.	3	Understanding
Contents: Navigation - methods of navigation - Radio direction finder - loop antenna – goniometer - Radio Navigation systems - ADF / NDB – Radio compass ADF, VHF phase comparison using ADF - Hyperbolic navigation systems – basic principle - LORAN, Omega, DECCA - Radio ranges – VOR – ground equipment – VOR receiver - Doppler VOR – DME.			
CO4	Understand the two types of landing systems and satellite navigation systems		
M4.01	To explain the function of components in an Instrument Landing System (ILS) with the help of diagram and significance of glide slope and markers	3	Understanding
M4.02	Explain Microwave Landing System (MLS)	3	Understanding
M4.03	State the basic principle of INS and DMS	3	Understanding
M4.04	Explain the basic ideas of GPS and DGPS	2	Understanding
M4.05	To explain GNSS and its classification	3	Understanding

	Series Test – II	1	
Contents: Instrument Landing System (ILS) – elements – localizer - glide slope – marker beacons -lighting systems – operation – limitation - Microwave Landing System (MLS) – operation – advantages – disadvantages - Navigation systems – INS – DNS - Satellite Navigation systems – GPS – DGPS – GNSS – COMPASS – DORIS – GALILEO – IRNSS – QZSS (concept only)			

Text / Reference

T/R	Book Title/Author
T1	Introduction to Radar Systems – Merrill I. Skolnik – McGraw Hill
R1	Radar Systems and Radio Aids to Navigation – Dr. A. K. Sen, Dr. A. B. Bhattacharya KhannaPublishers
R2	Elements of Electronic Navigation – N. S. Nagaraja – Tata McGraw Hill
R3	Radar, Sonar and Navigation Engineering – K. K. Sharma – Katson Books